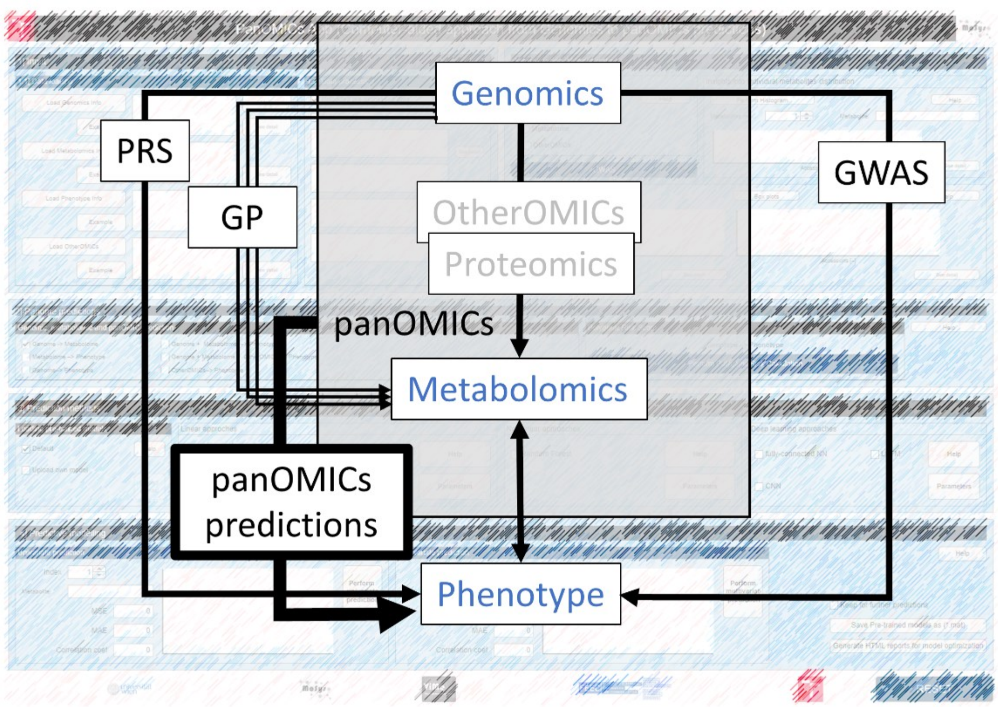
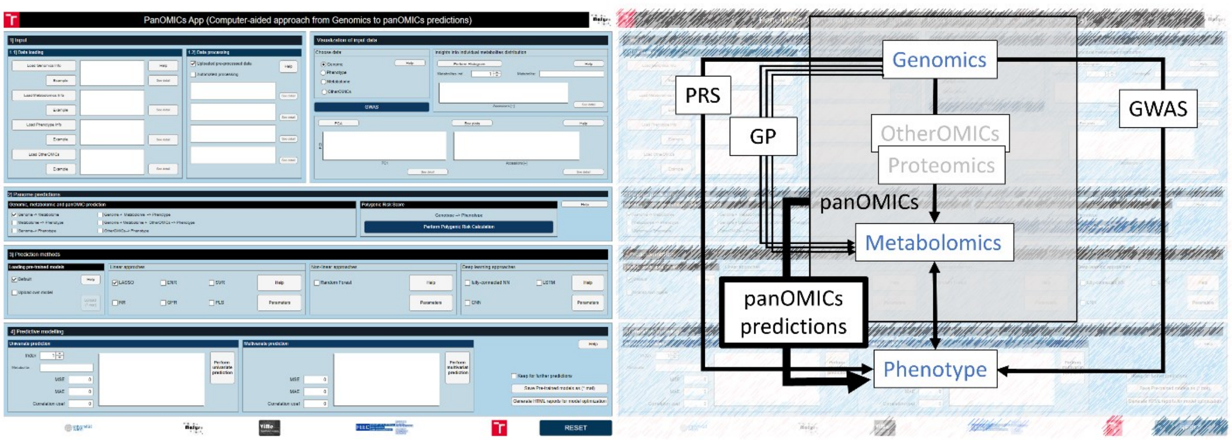
PANOMICS App Manual



Introduction

Advancements in sequencing technologies have propelled molecular biology into the post-genomic era, where research increasingly focuses on understanding the functional relationships between genetic variation and phenotypic outcomes.

The PANOMICS App, developed in MATLAB R2023b App Designer, provides an integrated and user-friendly environment for uploading, processing, visualizing, and analyzing genomic and metabolomic data. The application supports multiple prediction frameworks, including genomic, polygenic, and panOMICs approaches, and implements a range of linear, non-linear, and deep learning methods.



Installation

Option 1: MATLAB installed

If MATLAB is installed on your system, the application can be executed directly:

1. Navigate to:

    PANOMICSApp/for_testing/

2. Run:

    PANOMICs.exe

Option 2: Standalone version (no MATLAB required)

If MATLAB is not available or licensed:

1. Navigate to:

    PANOMICSApp/for_redistribution/

2. Run the installer:

    MyAppInstaller_web.exe

3. After installation, download the example input data from:

    PANOMICSApp/for_testing/input_example

4. Place the data in the same directory as the installed application.

Important (required for application launch):

Before running the standalone application, it is necessary to download and install the MATLAB Runtime R2023b (23.2).

The runtime is available for free and no MATLAB license is required o run the full desktop application.

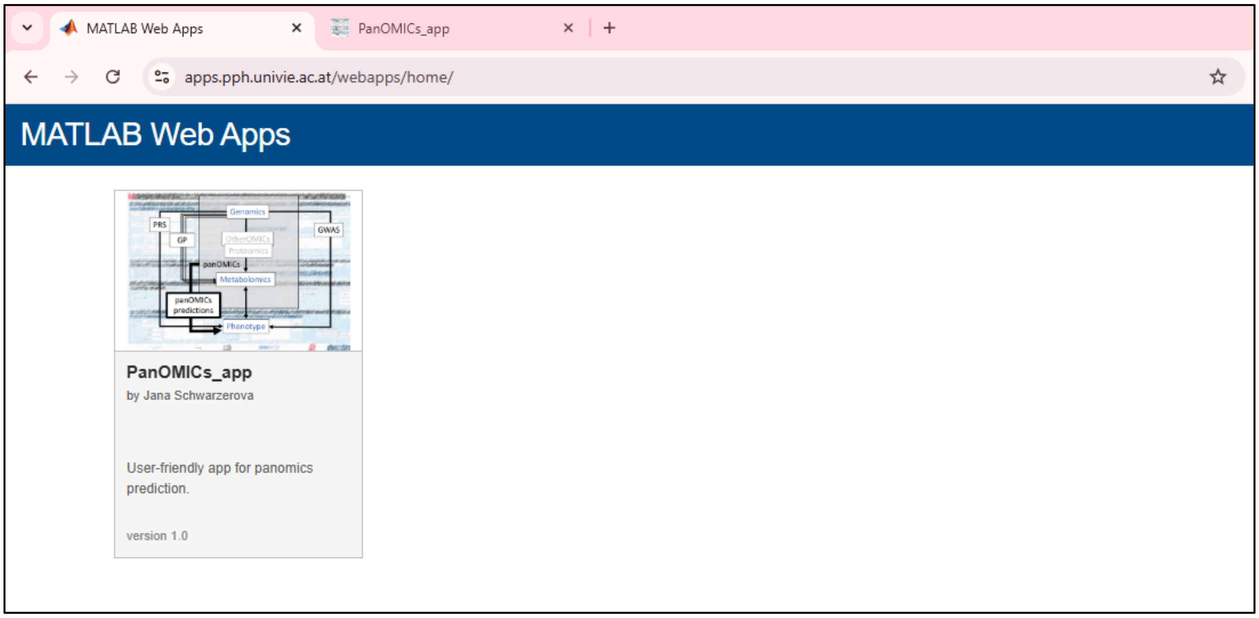
Download here:

https://www.mathworks.com/products/compiler/matlab-runtime.html

Install the runtime first, then proceed with the application installation.

Option 3: Web-based version

A web-based version of the PANOMICS App is also available for demonstration and lightweight analyses: https://apps.pph.univie.ac.at/webapps/home/

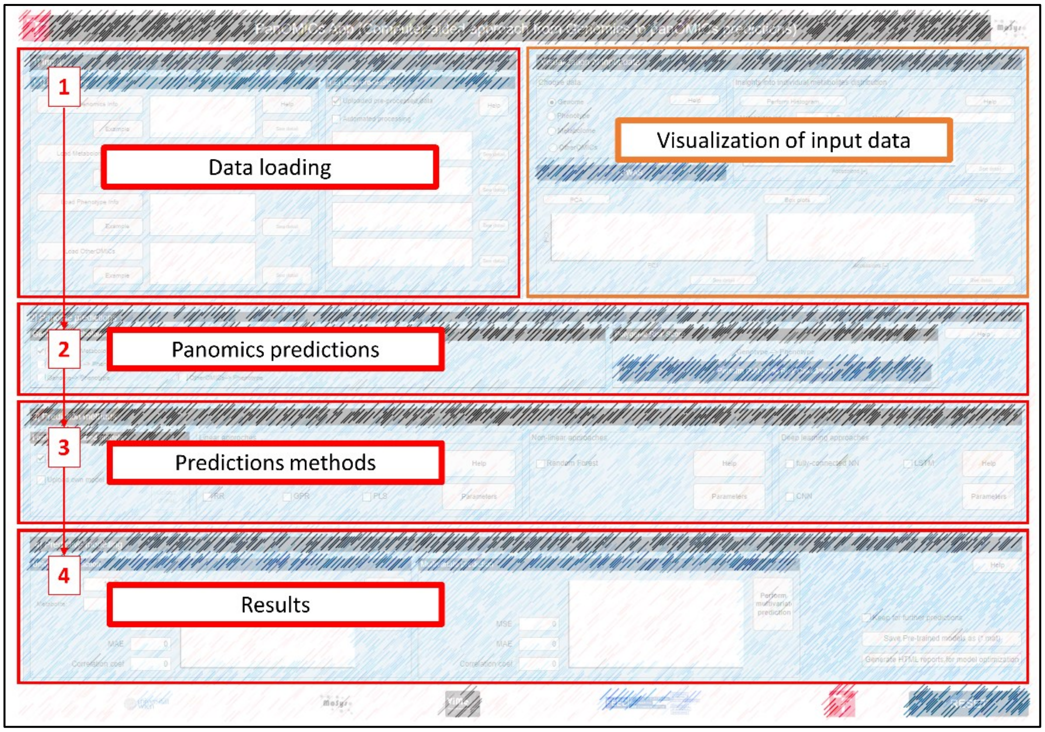


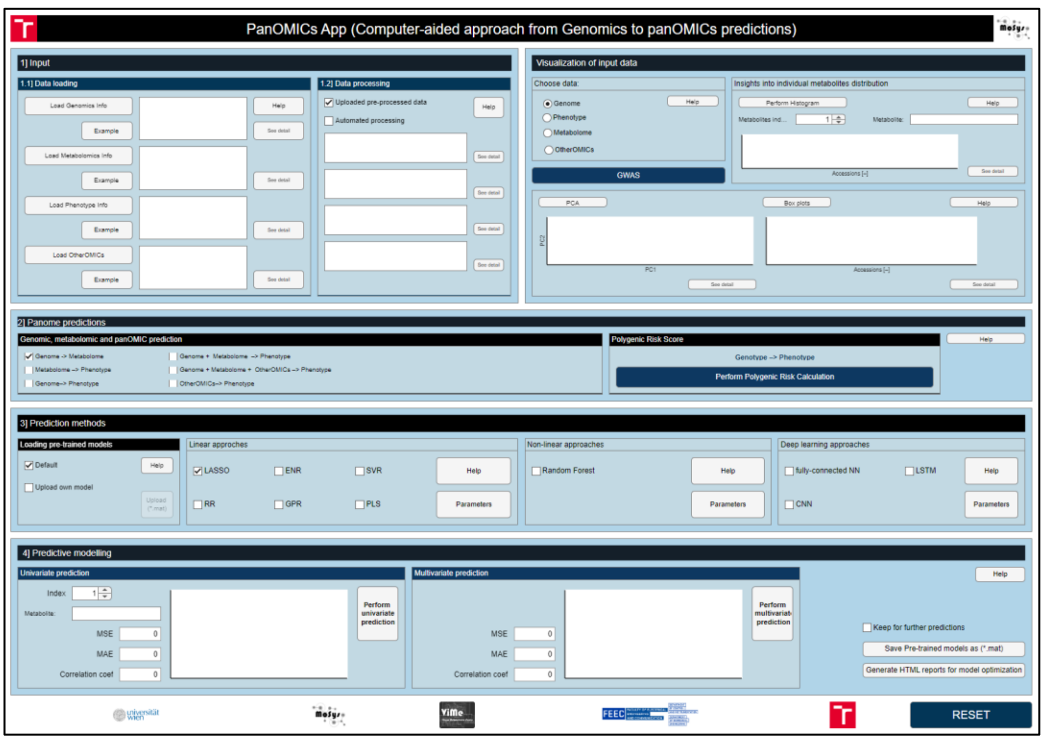
The web and desktop versions share a consistent interface and do not require a MATLAB license. Users can upload datasets, perform preprocessing, visualize data, and run prediction analyses directly through the browser.

The web version provides the full core functionality of the application; however, performance may be limited by server-side computational resources, particularly for large datasets or computationally intensive analyses. For such cases, the standalone desktop version is recommended.

Application Workflow and Functionality

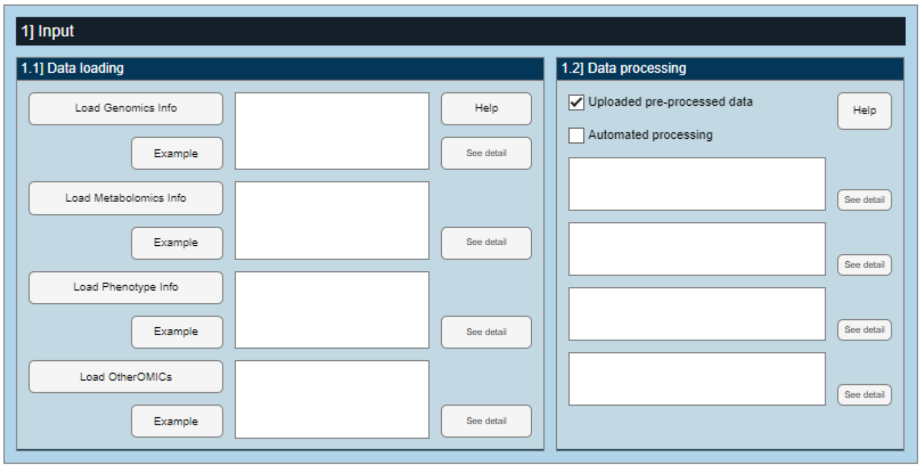
The PANOMICS App is organized into four main functional sections that guide users through data upload, exploration, prediction, and output management.



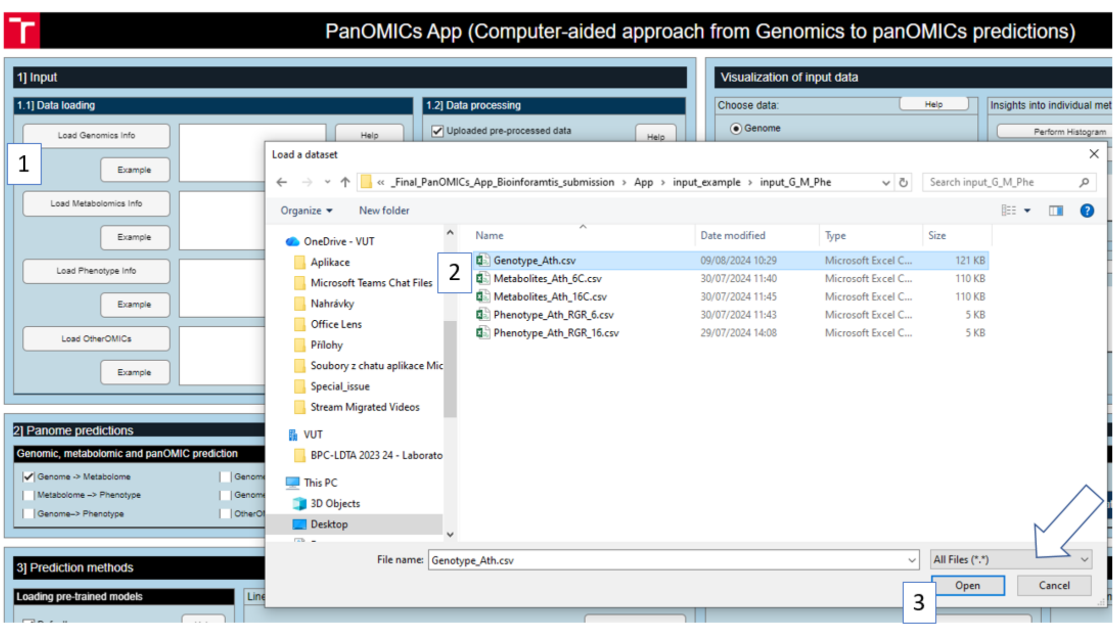


1] Input section

Input section of the app is designed for uploading data. The app expects genomic data in the form of SNP matrices, where '0' indicates an undetected SNP and '1' indicates a detected SNP. Users can also upload metabolomic data related to individual samples associated with the SNP data. Additionally, phenotypic data can be uploaded for calculating Polygenic Risk Scores (PRS) and panOMICs predictions.



Users can upload genomic datasets in SNP matrix format, optionally supplemented with metabolomic and phenotypic data. Pre-processed datasets can be analyzed directly, while raw data can be processed using an automated preprocessing pipeline. This includes imputation of missing values (replaced with zeros for genomic and phenotypic data), substitution of missing metabolomic values with half of the minimum detected value, and log10 transformation of metabolite intensities.

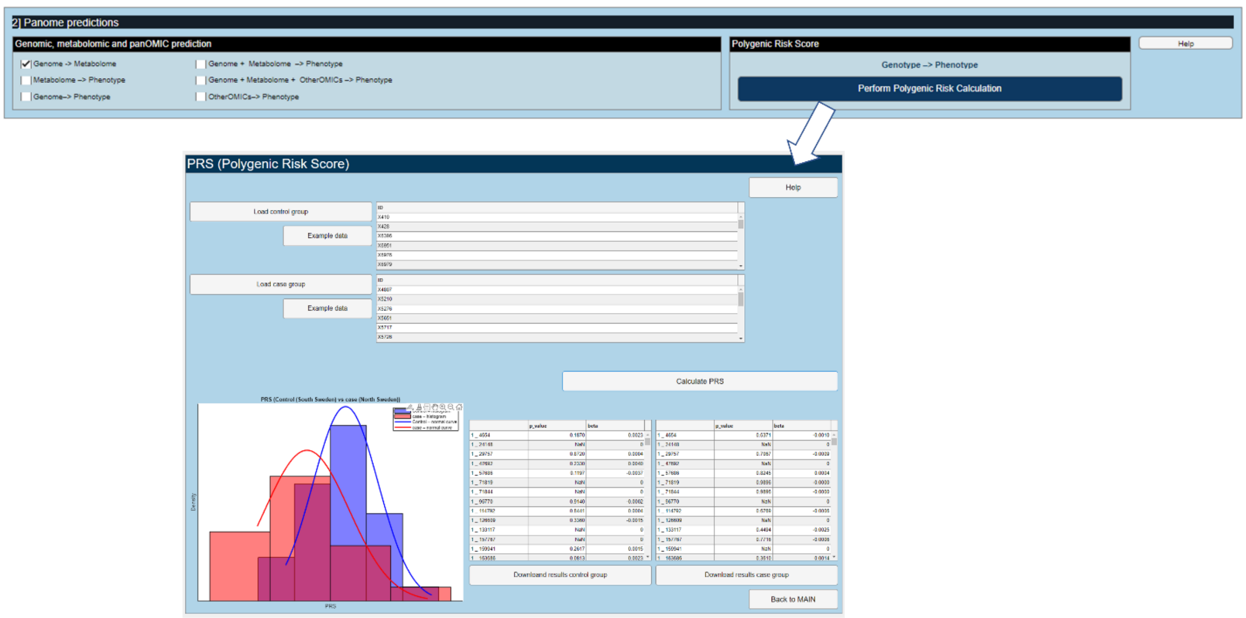


Visualization section

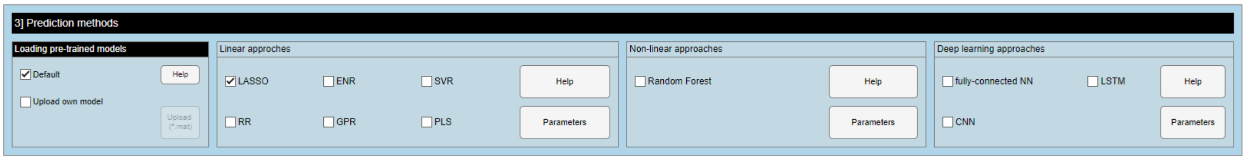
The Visualization module enables interactive exploration of input datasets. Available tools include principal component analysis (PCA), boxplots, and histograms for both genomic and metabolomic data distributions. In addition, genome-wide association study (GWAS) analysis can be performed for genomic datasets, providing a comprehensive overview of underlying genetic structure and variability.

2] Panome prediction section

The central functionality of the application is implemented in the Panome prediction section, where users define the overall prediction framework. This includes the selection of the prediction task, such as genomic prediction, metabolome prediction, other-omics integration, or polygenic risk score (PRS) estimation. The section provides a structured interface that links data input with downstream modelling and enables seamless transition to the prediction algorithms implemented in the Prediction Methods section.



3] Prediction methods section



The Prediction Methods section provides access to a broad range of machine learning and statistical models used for predictive analysis. Users can select the desired prediction task and apply one or more algorithms depending on the data type and research objective.

Supported models include:

Linear methods:

LASSO – Least Absolute Shrinkage and Selection Operator,

RR – Ridge Regression,

ENR – Elastic Net Regression,

GPR – Gaussian Process Regression,

SVR – Support Vector Regression,

PLS – Partial Least Squares,

Non-linear methods:

RF – Random Forest,

Deep learning approaches:

fully connected NN – fully connected Neural Networks,

LSTM – Long Short-Term Memory,

CNN – Convolutional Neural Network

PRS analysis requires SNP positional information and produces downloadable CSV output files accompanied by relevant visualizations. For non-linear and deep learning models, users can further optimize performance by adjusting model-specific hyperparameters via the Parameters option.

Linear and statistical models – detail description

LASSO (Least Absolute Shrinkage and Selection Operator) regression applies L1 regularization to enhance model performance by performing both variable selection and regularization. In this method, the model is trained to predict exam scores using the LASSO and elastic net methods. Predictions are compared with actual exam grades using the formula:

```
app.Pred_Val = XTest*coef + coef0;
```

RR (Ridge Regression) is a linear model incorporating L2 regularization to penalize large coefficients and mitigate multicollinearity. In the PANOMICS application, RR is implemented with a very small alpha parameter (e.g., 0.000001), approximating ridge regression. The model is trained using 5-fold cross-validation to find the optimal lambda value, enhancing stability and prediction accuracy.

ENR (Elastic Net Regularization) combines L1 and L2 penalties to leverage the strengths of both LASSO and Ridge Regression. In this approach, the alpha parameter is set to 0.75, balancing LASSO and Ridge contributions. The ENR model undergoes 5-fold cross-validation to determine the best regularization parameters, which are then used for predicting outcomes.

GPR – Fit a Gaussian process regression (GPR) model; GPR involves fitting a Gaussian Process Regression model using the fitrgp function. This model is trained on sample data, accommodating both univariate and multivariate outputs, and provides flexible prediction capabilities

gprMdl = fitrgp(Tbl,ResponseVarName) returns a Gaussian process regression (GPR) model trained using the sample data in Tbl, where ResponseVarName is the name of the response variable in Tbl.

(https://www.mathworks.com/help/stats/fitrgp.html)

SVR – Fit a support vector machine regression (SVR) model; SVR fits a support vector machine regression model to predictor data, utilizing kernel functions and soft-margin minimization techniques. The ***fitrsvm*** function supports various methods, including SMO and quadratic programming. For high-dimensional data, fitrlinear is used. For binary classification, fitcsvm and fitclinear are employed based on data dimensions

fitrsvm trains or cross-validates a support vector machine (SVM) regression model on a low- through moderate-dimensional predictor data set. fitrsvm supports mapping the predictor data using kernel functions, and supports SMO, ISDA, or *L*1 soft-margin minimization via quadratic programming for objective-function minimization.

To train a linear SVM regression model on a high-dimensional data set, that is, data sets that include many predictor variables, use fitrlinear instead.

To train an SVM model for binary classification, see fitcsvm for low- through moderate-dimensional predictor data sets, or fitclinear for high-dimensional data sets.

(https://www.mathworks.com/help/stats/fitrsvm.html)

PLS – Partial least-squares (PLS) regression – PLS regression calculates predictor and response loadings using the ***plsregress*** function, which identifies relationships between predictors and responses by specifying the number of components.

plsregress(X,Y,ncomp) returns the predictor and response loadings XL and YL, respectively, for a partial least-squares (PLS) regression of the responses in matrix Y on the predictors in matrix X, using ncomp PLS components.

(https://www.mathworks.com/help/stats/plsregress.html)

Non-linear methods – detail description

***RF* –** Random forest implemented by **oobQuantilePredict** - Quantile predictions for out-of-bag observations from bag of regression trees

(https://www.mathworks.com/help/stats/treebagger.oobquantilepredict.html)

Deep Learning approaches:

***Fully connected Neural Network (NN)* –** A feedforward neural network is used for non-sequential representation learning.

Architecture:

Feature input layer => Multiple fully connected layers => ReLU activation layers => Regression output layer

This architecture is optimized for general-purpose prediction tasks without explicit sequence modeling.

***LSTM* –** LSTM networks are designed to model sequential dependencies in high-dimensional genomic data. SNPs are treated as ordered feature sequences, enabling the model to capture long-range interactions.

Architecture:

Sequence input layer => LSTM layer => Fully connected layer => Dropout layer => Fully connected output layer => Regression layer

Training is performed using the Adam optimizer with GPU acceleration when available. Mini-batch size, number of epochs, gradient threshold, and layer size are user-defined.

CNN – CNN models extract local patterns from genomic sequences using convolutional filters.

Architecture:

Sequence input layer =>Stacked 1D convolutional layers (causal and same padding) => ReLU activation layers => Fully connected layers => Regression output layer

This architecture enables hierarchical feature extraction from SNP sequences and metabolomic profiles.

All deep learning and non-linear models support user-defined hyperparameter tuning via the Parameters button in the graphical interface, where users can modify the settings..

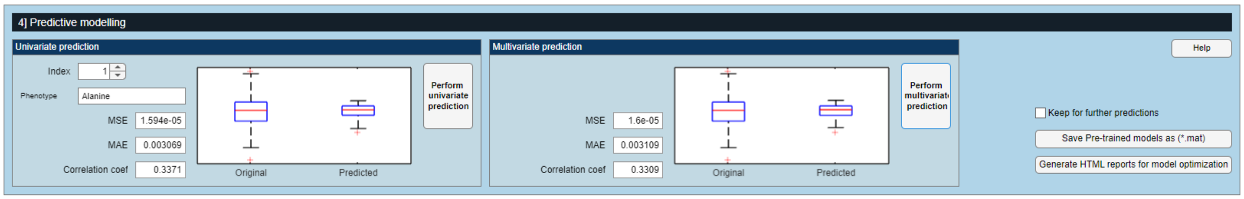
All deep learning models are trained using MATLAB’s trainingOptions function with the Adam optimizer. GPU acceleration is automatically enabled if available; otherwise, training is performed on CPU.

All models are integrated into a unified pipeline that ensures consistent preprocessing, training, and evaluation across methods.

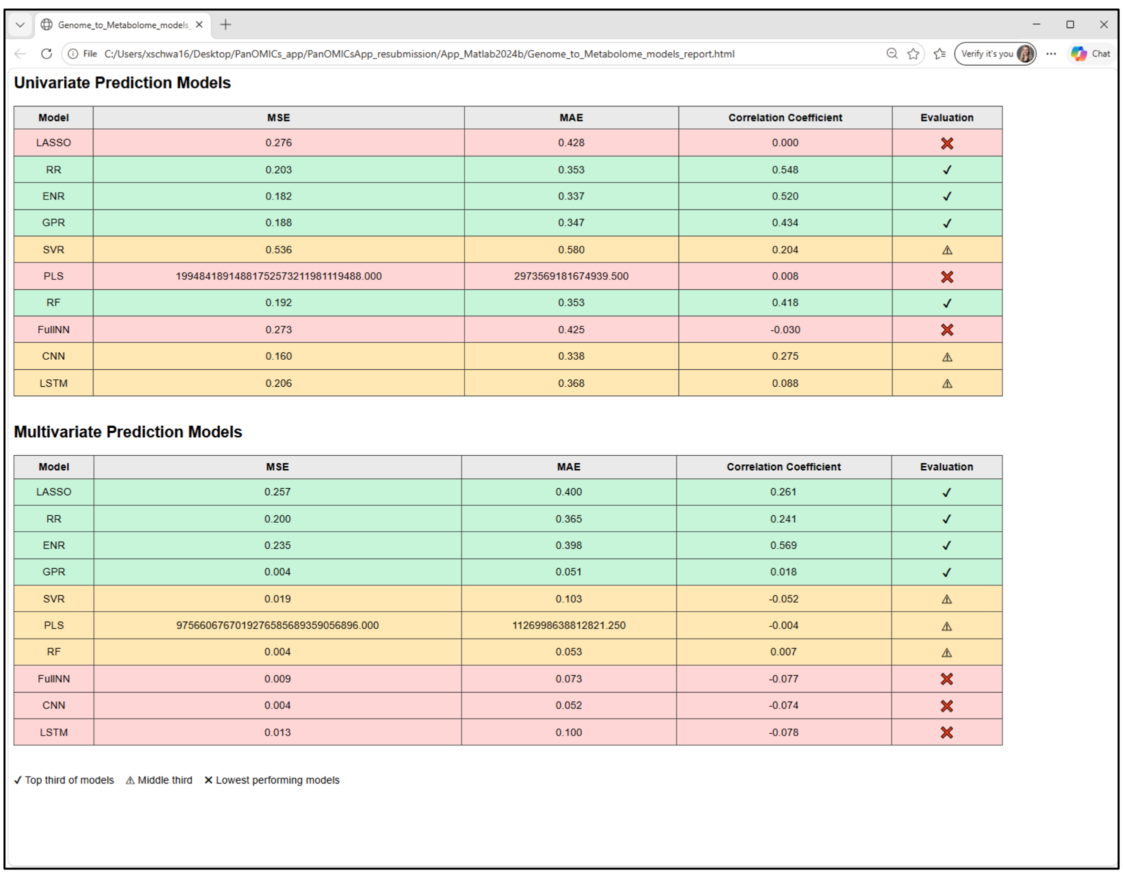
4] Predictive Modelling

The Output and Predictive Modelling section manages the presentation, evaluation, and storage of prediction results. Outputs are provided in two formats: univariate (individual metabolites or target variables) and multivariate (complete metabolite or feature sets).

Model performance is assessed using standard evaluation metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), and correlation coefficient (cc). In addition, predicted versus observed values can be visualized using boxplots for rapid assessment of model accuracy.



Results can be saved in *.mat* format within this section and later reloaded in the Input section for further analysis or integration with new datasets. To support model selection and benchmarking, the application also includes an option to generate HTML reports, enabling systematic comparison of prediction performance across all implemented methods.



Univariate Prediction Models

Model	MSE	MAE	Correlation Coefficient	Evaluation
LASSO	0.276	0.428	0.000	✗
RR	0.203	0.353	0.548	✓
ENR	0.182	0.337	0.520	✓
GPR	0.188	0.347	0.434	✓
SVR	0.536	0.580	0.204	⚠
PLS	19948418914881752573211981119488.000	2973569181674939.500	0.008	✗
RF	0.192	0.353	0.418	✓
FullNN	0.273	0.425	-0.030	✗
CNN	0.160	0.338	0.275	⚠
LSTM	0.206	0.368	0.088	⚠

Multivariate Prediction Models

Model	MSE	MAE	Correlation Coefficient	Evaluation
LASSO	0.257	0.400	0.261	✓
RR	0.200	0.365	0.241	✓
ENR	0.235	0.398	0.569	✓
GPR	0.004	0.051	0.018	✓
SVR	0.019	0.103	-0.052	⚠
PLS	9756606767019276585689359056896.000	1126998638812821.250	-0.004	⚠
RF	0.004	0.053	0.007	⚠
FullNN	0.009	0.073	-0.077	✗
CNN	0.004	0.052	-0.074	✗
LSTM	0.013	0.100	-0.078	✗

✓ Top third of models ⚠ Middle third ✕ Lowest performing models